

MID-TERM EXAMINATION

Course: **Logical Structures**

Time: 60 minutes

Term: 2 – Academic year: 2020-2021

Lecturer(s): **Bui Anh Tuan**

Student name:

Student ID:

(Notes: Neither books nor laptops, phones allowed)

This exam consists of 5 questions as follows:

Question 1) Verify the validity of the following sequent using natural deduction rules

$$(p \rightarrow q) \rightarrow (p \rightarrow r) \vdash (p \vee r) \rightarrow (q \rightarrow r)$$

Question 2) Suppose that a formula F has the following truth table

p	q	r	F
1	1	1	1
1	1	0	0
1	0	1	1
1	0	0	0
0	1	1	0
0	1	0	0
0	0	1	1
0	0	0	0

Provide a Conjunctive Normal Form of the formula F .

Question 3) Let $\phi = (p \rightarrow q) \rightarrow (p \vee q)$. Compute $\text{CNF}(\text{NNF}(\text{IMPL_FREE}(\phi)))$. Is ϕ valid?

Question 4) Apply algorithm HORN to the following Horn formula

$$(p_5 \rightarrow p_{11}) \wedge (p_2 \wedge p_3 \wedge p_5 \rightarrow p_{13}) \wedge (p_5 \wedge p_{11} \rightarrow \perp) \wedge (\top \rightarrow p_5)$$

Is it satisfiable?

Question 5) Let $\phi = ((p \vee q) \wedge (p \rightarrow r)) \rightarrow r$

- Draw the DAG of $T(\phi)$
- Use SAT solver to check the satisfiability of ϕ .

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